

The Augmented Synthetic Historical Control Method

Introduction

Much of economic and policy research which uses causal inference focuses on the idea of comparing a treated set of units to a set of control/donor units. The synthetic control method (SCM) of [Abadie et al. \[2010\]](#) is one of the most widely used approaches in panel data econometrics for estimating treatment effects, second only to difference-in-differences. The idea behind SCM is simple: the untreated potential outcomes of a treated unit can be approximated by a weighted average of donor pool units. Difference-in-Differences designs are largely based on a similar idea, under a parallel pre-intervention trends assumption [\[Baker et al., 2025\]](#).

In many cases however, no appropriate donors exist—such as during global events (e.g., pandemics, financial crises) or when donor days simply does not exist. To address this, [Chen et al. \[2024\]](#) introduced the synthetic historical control method (SHC). Rather than relying on external (usually geographic) donors, SHC uses the treated unit’s own pre-treatment history as a donor pool by splitting the pre-treatment period into overlapping segments. A quadratic program then estimates weights over these segments to predict counterfactual outcomes, predicated on the idea that future outcomes resemble past patterns. SHC, like [Abadie et al. \[2010, 2015\]](#), restricts weights to the simplex, ensuring convex combinations of donor segments. This guarantees interpretability but can be too restrictive in time-series settings where noise or complex dynamics make exact pre-treatment fit unattainable. In such cases, purely convex weights may not provide accurate approximations.

This paper introduces the *Augmented Synthetic Historical Control* method (ASHC). ASHC extends SHC by [Chen et al. \[2024\]](#) in two key ways. First, it relaxes the simplex constraint to an affine constraint, requiring only that weights sum to one. This allows for negative weights, permitting extrapolation beyond the convex hull of observed segments in exchange for better fit. Second, it adds a quadratic proximity penalty that discourages large deviations from the SHC solution, in the spirit of the augmented SCM of [Ben-Michael et al. \[2021\]](#), which encourages the estimator to standard SCM unless extrapolation is needed. ASHC retains, however, the stabilizing Gram penalty from [Chen et al. \[2024\]](#), which regularizes low-variance directions in the donor matrix. Together, these three components yield a new estimator that generalizes SHC to settings of imperfect pre-treatment fit.

We establish the theoretical properties of ASHC. We prove a finite-sample error bound, decomposing the error into four interpretable parts: SHC residual error, a regularization-controlled deviation term, smoothness-induced bias, and noise. We then show that ASHC is asymptotically consistent: as pre-treatment length and smoothing order grow, prediction error converges to the noise floor, and under sub-Gaussian noise, to zero in probability. Finally, we derive the convergence rate, finding that under the optimal bandwidth, the error decays approaches a near-parametric rate. Together, these results demonstrate that ASHC preserves the interpretability of SHC while offering provably stronger error control and asymptotic guarantees.

Section 2 introduces the assumptions and implementation of the SHC method. Then, I introduce the ASHC method. Section 3 presents the finite sample and asymptotic analysis. Section 4 presents calibrated simulations. Section 5 presents an empirical application. All proofs are presented in the appendix. The implementation of this was carried out using Python’s `mlsynth` library (<https://mlsynth.readthedocs.io/en/latest/>).

References

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